Predicting Residual Cancer Burden from Pre-Treatment Breast Cancer Biopsy Pathology Images

Team 1

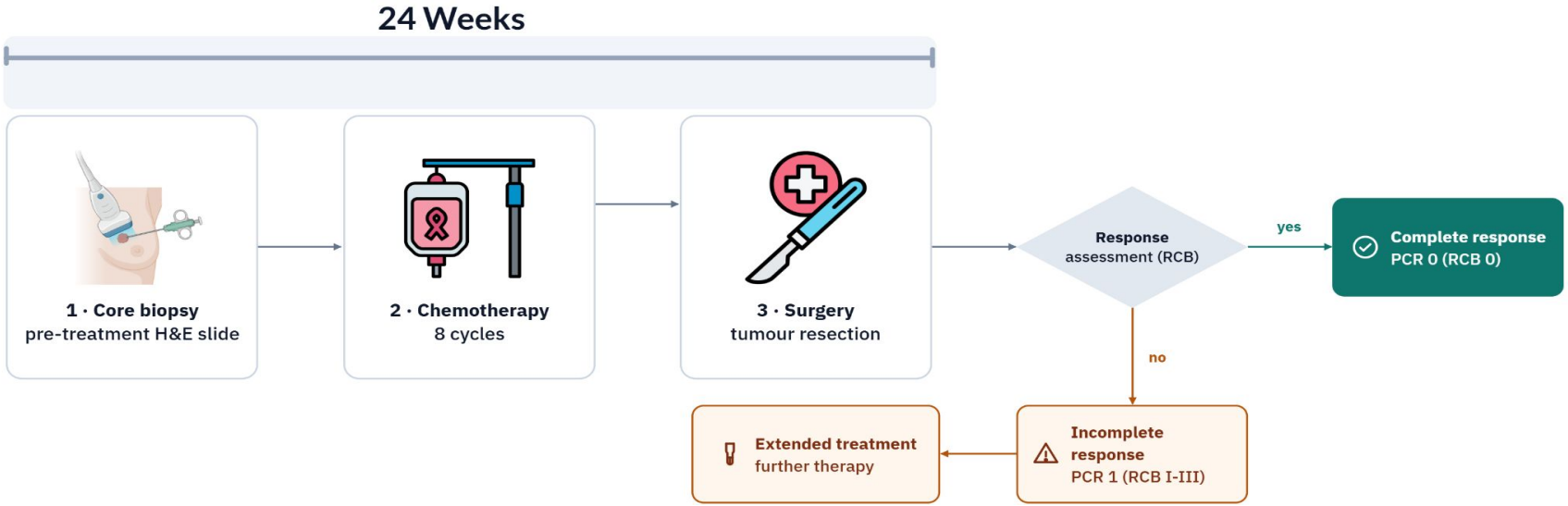
2026 National Health Datathon

Breast cancer patients can have a taxing treatment process

Patients with locally advanced breast cancer undergo months of chemotherapy prior to surgery with varying levels of treatment response

Patients with incomplete response may undergo intensified cancer treatments

Predicting response at time of biopsy can inform patients decisions on treatment trajectory



Diagnosing breast cancer samples

High-resolution digital pathology imaging of breast cancer dataset

- n=107 (patients with Biopsy Pathology Images **and** a recorded outcome)

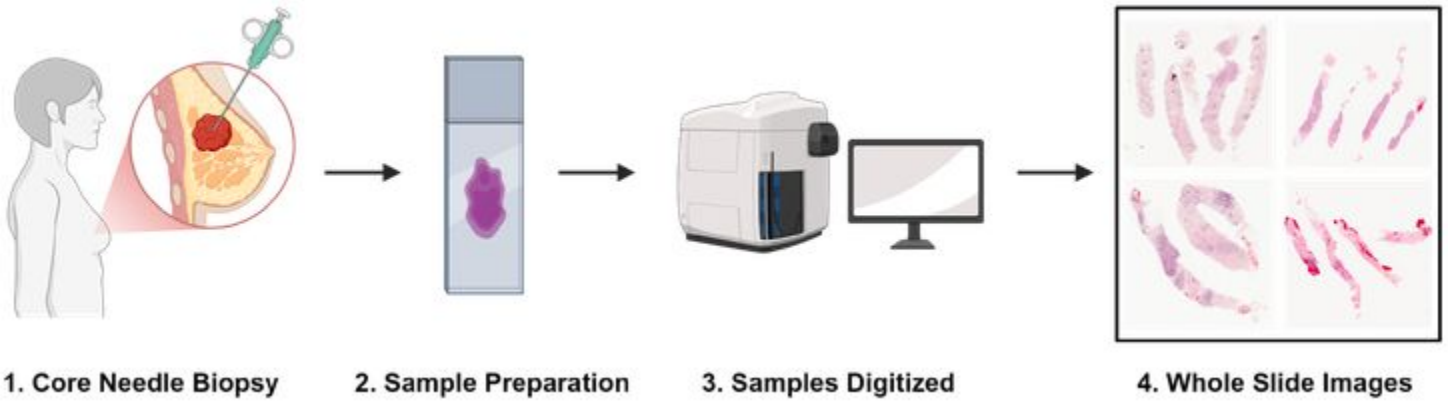
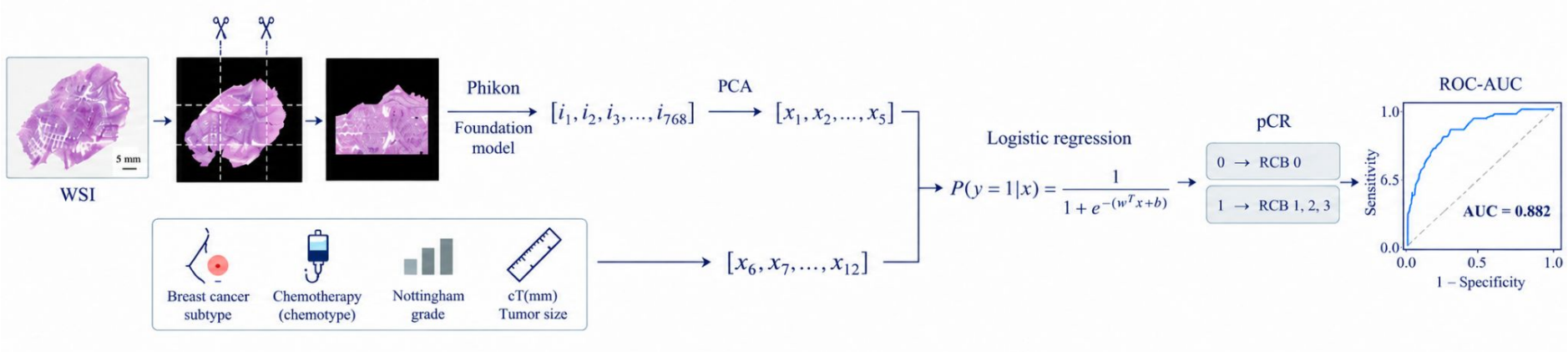


Image from T-CAIREM, Matt Bielecki

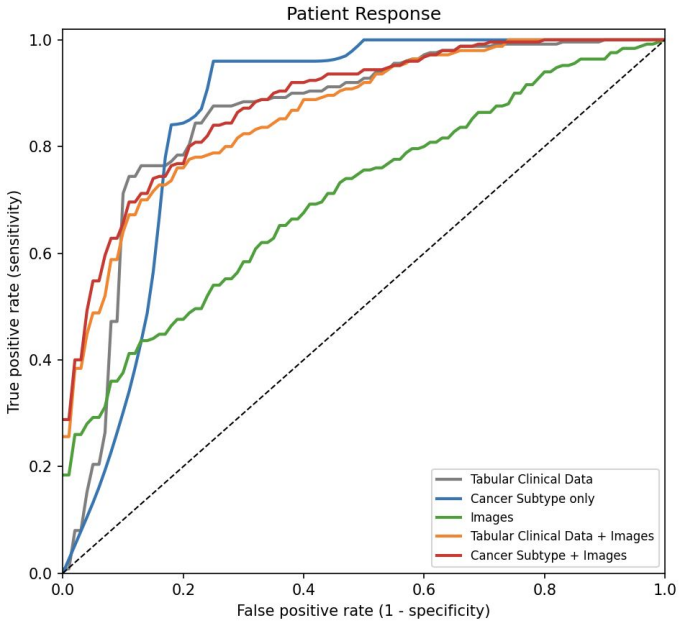
Predicting patient response from pathological imaging

Research question: can pathological whole-slide images in conjunction with other clinical factors (e.g. ER+/-) from breast cancer biopsies be used to predict pathological complete response (pCR)?



Foundational pathology model predicts patient response

Inclusion of cancer subtype data with pathological images yields best model performance



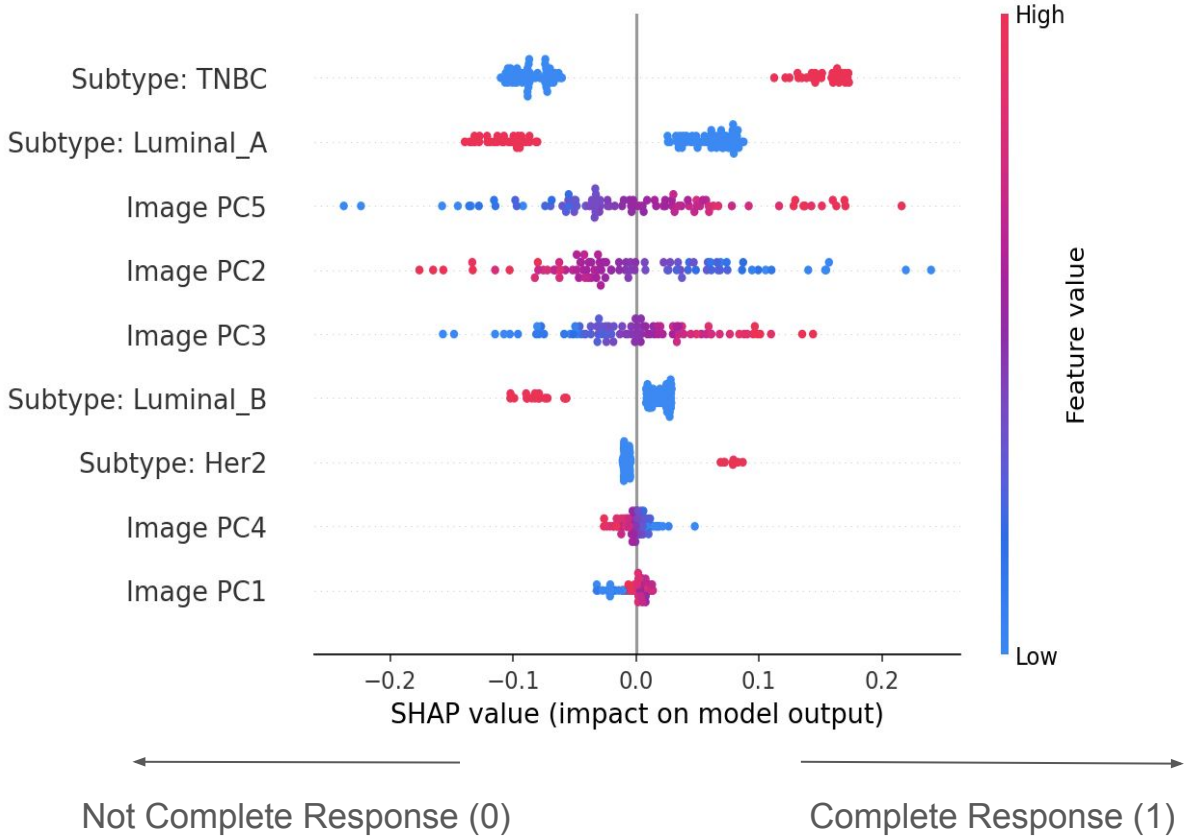
Model	ROC-AUC
Tabular Clinical Data	0.856
Cancer Subtype only	0.858
Images	0.704
Tabular Clinical Data + Images	0.862
Cancer Subtype + Images	**0.882**

SHAP analysis demonstrates feature contributions

SHAP values: show contributions of variables to outcome predictions

Cancer subtype contribution correlates with response and matches clinical expectations

Image features are informative for complete response prediction



Conclusions and future directions

A foundation open source model can predict patient response from clinical variables and pathological images

We can provide more informative decisions for patients and doctors at time of biopsy

While we obtained a high predictive value (AUC 0.882), the model could be improved with additional training and external datasets

Approach	Studies · reported pCR AUC	Novelty with our project
Feature identification segment, then measure TILs, stroma, nuclei	Saednia 2022[1], Huang 2023 (IMPRESS)[2], Aswolinskiy 2023 (PROACTING)[3] · 0.59–0.91	No manual annotation required. The whole biopsy is embedded automatically.
ImageNet CNNs networks trained on everyday photos	Li 2021[4], Li 2022[5], Zeng 2024[6], Sturm 2025[7] · 0.70–0.85	We use a model trained on pathology (40 million H&E tiles), not on general photos.
Small self-supervised / hybrid small ViT fused with a CNN	Valderrama 2026[8], Lin 2026[9] · 0.65–0.81 external	We use one encoder on its own, so the image signal has a single, clear source.
Foundation model + MIL gated UNI with attention pooling	Khan 2025[10], Hou 2026[11], Zhou 2026[12] · 0.63–0.82 external	We use Phikon: open weights, public training data, and not used in any prior response study. Five components keep the model to 12 inputs or fewer for 107 patients.
Our approach NHD 2026	Phikon[13] embeddings → 5 components + subtype → logistic regression · **0.882**	Subtype alone 0.858; images alone 0.704. n = 107, public Health Data Nexus dataset[14]. No annotation; anyone can reproduce the features.

References

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